Mathematics as a Foreign Language

Errata

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These corrections refer to the book

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D4	`https://doi.org/10.1007/978-3-032-30721-7_9`

1 Corrections of mathematical statements

Page 187 Chap. C1, Sect. C1.2, paragraph beginning "All the stated propositions are weaker".

For: "…and the theorem that a countable union of countable sets is countable is equivalent to countable choice from countable sets."

Read: "…and the theorem that a countable union of countable sets is countable implies countable choice from countable sets, but not conversely [P. Howard, The axiom of choice for countable collections of countable sets does not imply the countable union theorem, *Notre Dame J. Formal Logic* 33(2) (1992), 236–243]."

Comment: To prove the countable union theorem one must choose an enumeration of each of the sets, not merely an element of each. The Remark on p. 385 (see below) states the relation in the correct direction.

Page 209 Chap. C1, Sect. C1.4.3, paragraph beginning "Now let us consider the model $V_\omega \setminus \{\varnothing\}$".

For: the whole paragraph.

Read: delete the paragraph.

Comment: The structure $\langle V_\omega \setminus \{\varnothing\}, \in\rangle$ does not have the stated properties. Extensionality fails: the distinct sets $\{\{\varnothing\}\}$ and $\{\varnothing, \{\varnothing\}\}$ both belong to the carrier and have in it the same single element $\{\varnothing\}$. Nor is the empty set absent: the element $\{\varnothing\}$ has no elements in the carrier and therefore plays the role of the empty set. In general, if membership is well-founded on the carrier, any $\in$-minimal element of the carrier is an empty set of the model.

Page 216 Chap. C1, Sect. C1.4.3, paragraph beginning "Even with set theory itself".

For: "(recall the models of Quine and $V_\omega \setminus \{\emptyset\}$)"

Read: "(recall Quine's model)"

Comment: See the correction to p. 209.

Page 285 Chap. D2, Exercise D2.16.

For: "(Hint: Consider a model consisting of two copies of $\mathbb{N}$, one following the other)."

Read: "(Hint: Consider a model consisting of a copy of $\mathbb{N}$ followed by a copy of $\mathbb{Z}$.)"

Comment: The order $\mathbb{N}+\mathbb{N}$ is not a model of the theory of $\langle \mathbb{N}, < \rangle$: in $\langle \mathbb{N}, < \rangle$ every element other than the least one has an immediate predecessor, whereas the least element of the second copy has none. The answer to D2.16 uses the correct model $\mathbb{N}+\mathbb{Z}$.

Page 328 Chap. D3, Sect. D3.5, paragraph beginning "The power of such a 'leap'".

For: "…we can with one formula $\varphi(n, q)$, where q is a rational number, encode a convergent sequence of rational numbers. Real numbers that are the limits of such sequences are called **computable**."

Read: "…we can with one Σ_1-formula $\varphi(n, q)$, where q is a rational number, define a sequence of rational numbers q_n (the graph of the function $n \mapsto q_n$), and with another Σ_1-formula a modulus of convergence, i.e., a function $N(k)$ such that $|q_m - q_{m'}| \leq 2^{-k}$ for all $m, m' \geq N(k)$. Real numbers that are the limits of such sequences are called **computable**."

Comment: A Σ_1-formula defining the graph of a total function defines a recursive function; hence the corrected definition is the standard one. Both conditions are needed: a sequence of rational numbers with a Σ_1-definable graph that is increasing and bounded may have a non-computable limit (E. Specker, 1949), and for an arbitrary formula φ the limits are exactly the arithmetical real numbers.

Page 328 Chap. D3, Sect. D3.5, next paragraph.

For: "…for example, the intermediate value theorem, the extreme value theorem for a continuous function on a closed interval, the identity $e^{x+y} = e^x \cdot e^y$ …"

Read: "…for example, the intermediate value theorem, the identity $e^{x+y} = e^x \cdot e^y$ …"

Comment: There exist computable continuous functions on $[0, 1]$ that attain their maximum at no computable point; hence the extreme value theorem does not hold for functions considered on computable points.

Page 379 Chap. D4, Sect. D4.3.4, paragraph following the definition of τ^+.

For: the sentences from "Here the first problems with cardinals begin." to "So in the theory ZFC, cardinals form a proper class."

Read: "The existence of τ^+ for every cardinal τ is provable already in ZF: one may take for τ^+ the Hartogs number $\mathfrak{h}(\tau)$, i.e., the least ordinal that cannot be injectively embedded into τ (Sect. D4.4.2). In particular, the cardinals form a proper class. The difficulties with cardinals in ZF lie elsewhere: ZF does not prove that the set $\mathscr{P}(\tau)$ can be well-ordered, i.e., that it has a cardinal as its cardinality. By adding the Axiom of Choice, with the help of Zermelo's theorem, we can guarantee the well-ordering of any set, and thus the existence of a cardinality for that set."

Page 379 Chap. D4, Sect. D4.3.4, Theorem D4.9.

For: "Let $\mathscr{K}$ be a proper class, well-ordered by some class-relation."

Read: "Let $\mathscr{K}$ be a proper class, well-ordered by some class-relation in such a way that every proper initial segment of $\mathscr{K}$ is a set."

Comment: Without this condition the statement fails: the class On, reordered so that 0 follows all other ordinals, is a well-ordered proper class with a greatest element and therefore is not isomorphic to $\langle \mathrm{On}, \in \rangle$.

Page 391 Chap. D4, Sect. D4.5, first paragraph.

For: "The fundamental result is that these two theories are mutually interpretable and, moreover, bi-interpretable, which means …"

Read: "The fundamental result is that these two theories are mutually interpretable, and after the axiom asserting the existence of transitive closures is added to HF, they are bi-interpretable, which means …"

Comment: See the correction to p. 392: the theory HF as defined in Chap. B2 does not prove the existence of transitive closures, and this axiom is required for bi-interpretability with PA.

Page 391 Chap. D4, Sect. D4.5, paragraph "Interpretation of PA in HF".

For: "All the necessary constructions (the concepts of the empty set, pair, union, transitive closure) are already available in HF."

Read: "All the necessary constructions (the concepts of the empty set, pair, union) are already available in HF."

Page 392 Chap. D4, Sect. D4.5.

For: "This mutual interpretability is not an isolated case. PA is also bi-interpretable with the theory Z-Inf (Zermelo's theory without the axiom of infinity)."

Read: "The precise form of this result is due to Kaye and Wong [R. Kaye, T. L. Wong, On interpretations of arithmetic and set theory, *Notre Dame J. Formal Logic* 48(4) (2007), 497–510]: PA is bi-interpretable with the theory ZF in which the Axiom of Infinity is replaced by its negation and the axiom asserting the existence of transitive closures is added. The negation of the Axiom of Infinity is essential: Zermelo's theory from which the Axiom of Infinity is merely deleted has models containing infinite sets (for example, $V_{\omega+\omega}$) and is not bi-interpretable with PA."

2 Further corrections

Page 112 Chap. B1, Sect. B1.2.7, "Group theory".

For: "…we should consider the multiplication of positive integers (the universe $\mathfrak{m}$) modulo a prime $\mathfrak{p}$ …"

Read: "…we should consider the multiplication of positive integers not divisible by a prime $\mathfrak{p}$ (the universe $\mathfrak{m}$) modulo $\mathfrak{p}$ …"

Comment: Multiples of $\mathfrak{p}$ have no multiplicative inverse modulo $\mathfrak{p}$.

Page 114 Chap. B1, Sect. B1.2.7, "Divisible Torsion-Free Abelian Groups".

For: "This excludes, for example, cyclic groups, …"

Read: "This excludes, for example, finite cyclic groups, …"

Comment: The group $\mathbb{Z}$ is cyclic and torsion-free.

Page 114 Chap. B1, Sect. B1.2.7, same subsection, last paragraph.

For: "Their complete and simple classification as $\mathbb{Q}$-vector spaces …"

Read: "Their complete and simple classification (as direct sums of copies of $\mathbb{Q}$ and of the quasicyclic groups $\mathbb{Z}(p^\infty)$) …"

Comment: Only torsion-free divisible groups are $\mathbb{Q}$-vector spaces.

Page 134 Chap. B2, Sect. B2.1.1, footnote on the axioms PA6.

For: "Therefore, multiplication can be defined through addition, but the totality and correctness of such a definition cannot be proven, whereas exponentiation and all higher recursions are definably correct in the presence of multiplication."

Read: "Multiplication cannot be defined by a formula in terms of addition alone: the theory of the natural numbers with addition (Presburger arithmetic) is decidable. The recursive definition of multiplication through addition remains external, since without multiplication the totality and correctness of such a definition cannot be proven, whereas exponentiation and all higher recursions are definably correct in the presence of multiplication."

Page 165 Chap. B2, Sect. B2.3.2.

For: "Abraham Fraenkel and Thoralf Skolem introduced the axiom of regularity and the axiom of replacement in the early 1920s, …"

Read: "The axiom of replacement was introduced by Abraham Fraenkel and Thoralf Skolem in the early 1920s, and the axiom of regularity by John von Neumann (1925) and Ernst Zermelo (1930), …"

Page 263 Chap. D2, end of Sect. D2.1.

For: "…every Boolean algebra is isomorphic to an algebra that is some (Cartesian) product of copies of the two-element Boolean algebra $\mathbf{2}$."

Read: "…every Boolean algebra is isomorphic to a subalgebra of some (Cartesian) product of copies of the two-element Boolean algebra $\mathbf{2}$."

Comment: Every power $\mathbf{2}^I$ is complete and atomic, whereas, for example, the countable atomless Boolean algebra is neither.

Page 274 Chap. D2, Sect. D2.2.3.

For: "…each subsequent constructible universe is built as the set of all *definable* subsets of the previous one, which means that in reality (from the point of view of the meta-theory) the cardinalities of all these universes are no more than countable."

Read: "…each subsequent constructible universe $L_{\alpha+1}$ is built as the set of all *definable* subsets of the previous one, so that L_α is countable for every countable ordinal α (from the point of view of the meta-theory); if ZF has a transitive model, then L_δ is a model of ZF for some countable ordinal δ."

Comment: $|L_\alpha| = |\alpha|$ for every infinite ordinal α.

Page 280 Chap. D2, Sect. D2.2.4, paragraph following the Łoś–Vaught test.

For: "…a first-order theory categorical in some infinite cardinality is not Gödelian."

Read: "…a first-order theory without finite models that is categorical in some infinite cardinality is not Gödelian."

Comment: The Łoś–Vaught test, on which the argument rests, requires the absence of finite models.

Page 283 Chap. D2, Table D2.2, footnote *.

For: "…by means of an axiom that specifies the characteristic of the field — 0 or a prime p."

Read: "…by means of an axiom that specifies the characteristic of the field — 0 or a prime p; it is these complete theories ACF_0 and ACF_p that are categorical in every uncountable cardinality, while ACF itself is categorical in no cardinality."

Comment: Algebraically closed fields of different characteristics and the same cardinality are not isomorphic.

Page 331 Chap. D3, Recommended Reading.

For: "The article by Kirby and Paris [40] became a landmark, presenting the first 'natural' combinatorial statement (a variation of Ramsey's theorem) independent of Peano's axioms."

Read: "The article by Kirby and Paris [40] became a landmark, presenting simple 'natural' combinatorial statements (Goodstein's theorem and the Hydra game) independent of Peano's axioms; the first statement of this kind, a variation of Ramsey's theorem, had been found a few years earlier by Paris and Harrington."

Page 339 Chap. D4, Sect. D4.1, paragraph following axioms BA1–BA7.

For: "…1 is the set M …"

Read: "…1 is the set U …"

Comment: $M = \mathscr{P}(U)$ is the carrier of the algebra; its greatest element is U.

Page 385 Chap. D4, Sect. D4.4.1, Remark following the theorem on countable unions.

For: "It turns out to be equivalent to a weaker choice axiom: the Axiom of Countable Choice for families of *countable* sets."

Read: "It implies, in turn, a still weaker choice principle, the Axiom of Countable Choice for families of *countable* sets: once the union $\bigcup_n A_n$ is enumerated, one may choose from each A_n its element with the least number."

Page 403 Answers to Exercises, A2.12, item 1.

For: "Transitive (brother of a brother is a brother)."

Read: "Not transitive in the strict sense: if a is a brother of b, then b is a brother of a, but a is not a brother of himself (for pairwise distinct persons, a brother of a brother is a brother)."